

Progress in IS

Ken Huang *Editor*

Agentic AI

Theories and Practices

Foreword by Matthew R. Versaggi and Yang Wang

 Springer

Progress in IS

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Ken Huang
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Agentic AI

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DistributedApps.ai

Fairfax, VA, USA

ISSN 2196-8705

ISSN 2196-8713 (electronic)

Progress in IS

ISBN 978-3-031-90025-9

ISBN 978-3-031-90026-6 (eBook)

<https://doi.org/10.1007/978-3-031-90026-6>

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To the pioneers who dared to dream of intelligent, autonomous systems, your vision has illuminated the path toward agentic AI, where machines act not only as tools but as collaborative partners reshaping our understanding of intelligence and agency.

To the researchers and practitioners exploring the complexities of agentic AI, your relentless pursuit of knowledge and innovation drives this transformative field forward, bridging theoretical possibilities with real-world applications.

To the organizations and policymakers working to harness the power of agentic AI for societal good, your commitment to creating frameworks that promote fairness, security, and accountability ensures that this technology benefits humanity as a whole.

To my colleagues and collaborators in the AI security communities such as Cloud Security Alliance, NIST, OWASP, SANS, and ISO, your insights, challenges, and support have been instrumental in crafting this book. It is a reflection of the shared quest to demystify agentic AI and make it accessible to all who seek to understand and apply its principles.

Lastly, to the readers of this book—innovators, students, and visionaries—your curiosity and ambition are what will shape the future of agentic AI. May this book serve as a catalyst for your endeavors, inspiring you to explore the profound potential of intelligent agents and their role in a rapidly evolving world.

Foreword



The journey of artificial intelligence has been one of profound transformation, evolving from abstract theoretical constructs to tangible tools that now shape industries, economies, and everyday lives. As an AI SME, a Distinguished Engineer, White House Presidential Innovation Fellow and someone deeply entrenched in the intersection of AI, healthcare, and quantum computing, I have witnessed firsthand the sweeping impact of AI-driven systems and the paradigm shifts they enable. Yet, amidst this progress, there lies a need for clarity, structure, and guidance for understanding and harnessing these transformative tools effectively. This is precisely what makes *Agentic AI: Theories and Practice* a timely and invaluable resource.

In this book, the author provides an excellent exploration of AI agents, their architecture, and their application across a diverse array of domains. Beginning with foundational concepts in Chap. 1, readers are introduced to the genesis and evolution of AI agents, accompanied by an engaging taxonomy and a glimpse into real-world examples that contextualize their practical significance. The meticulous organization of the book ensures that both novice and seasoned AI enthusiasts will

find value, whether seeking an introduction to multi-agent coordination or exploring advanced topics such as AI-enabled offensive and defensive security applications.

What sets this work apart is its balanced approach, seamlessly blending technical depth with practical insights. The chapters on AI agent tools, frameworks, and the seven-layer architecture offer a robust foundation for building and deploying AI agents effectively, while discussions on challenges provide a pragmatic view of the real-world obstacles developers and organizations may face. The exploration of multi-agent systems in Chap. 3 showcases the complexities of coordination, communication, and conflict resolution, while also highlighting the immense potential of these systems in solving global-scale problems.

Equally compelling is the book's attention to emerging domains. From AI agents in the agentic economy, where blockchain and tokenization redefine value exchange, to their role in reimagining workflows, offensive security, and healthcare, the breadth of application is truly remarkable. These chapters not only provide actionable insights but also inspire readers to envision possibilities beyond the current technological landscape.

The author's inclusion of sector-specific applications, such as in banking, insurance, and robotics, further emphasizes the universal relevance of AI agents. These industries, often regarded as traditional or slow to adopt innovation, are now poised to be revolutionized by agentic AI. Furthermore, the book's commitment to addressing safety and governance considerations in its final chapters underscores the importance of developing AI systems responsibly. As we march toward a future increasingly driven by autonomous systems, these discussions on ethical deployment and security are not just necessary; they are imperative.

In *Agentic AI: Theories and Practice*, the reader will find a comprehensive guide that bridges theory and practice, ensuring they are equipped to navigate this rapidly evolving field with confidence and purpose. This book is not just a reference but a call to action for innovators, researchers, and policymakers to collaborate and harness the transformative power of agentic AI responsibly and effectively.

Whether you are an AI practitioner, an entrepreneur seeking to integrate AI agents into your business, or a curious observer of this technological revolution, you will find this book to be a profound source of knowledge and inspiration. It is my honor to contribute this foreword and to commend the author for their exceptional effort in delivering a resource that will undoubtedly shape the future of agentic AI.

<https://www.linkedin.com/in/versaggi/>

Senior Leader in Artificial Intelligence Healthcare
and White House Presidential Innovation Fellow
in the Special AI cohort of '24
Washington, DC, USA

Matthew R. Versaggi

Foreword



The advent of generative AI has marked a profound shift in the way we perceive and interact with technology. Among the many facets of AI, the emergence of agentic AI systems has paved the way for a new paradigm, where autonomous, intelligent agents operate with increasing sophistication across diverse domains. As a mathematician who has explored both the theoretical and interdisciplinary applications of mathematics, I find this field deeply resonant with the evolving interplay between rigorous theory and practical innovation.

My own academic journey began at the University of Science and Technology of China and continued through my doctoral studies at Harvard University. Over the years, I have witnessed firsthand how advancements in mathematical models and computational techniques have laid the groundwork for the transformative technologies we see today. Mathematics has always been at the heart of artificial intelligence, serving as its structural backbone. The algorithms and models underpinning AI agents—be it optimization techniques, probabilistic reasoning, or machine learning—are all born of mathematical inquiry. As Vice-President for Institutional Advancement at the Hong Kong University of Science and Technology (HKUST), I also see the role of institutions as catalysts in bringing such innovations to life.

HKUST, with its emphasis on interdisciplinary research and global collaboration, embodies the spirit required to nurture breakthroughs in agentic AI.

“Agentic AI: Theories and Practices” is an ambitious and timely work that delves into the transformative world of AI agents. This book provides a comprehensive exploration of the field, from its theoretical genesis to its real-world implementations. Each chapter is meticulously structured to guide readers through foundational concepts, practical frameworks, and the implications of agentic AI across industries. For instance, the detailed discussions on multi-agent systems and their coordination, tokenized AI economies, and safety considerations are not only intellectually stimulating but also provide actionable insights for practitioners and policymakers alike.

One of the book’s most commendable aspects is its thorough examination of the dual use of AI agents and their integration with generative AI models, illustrating how these technologies can be harnessed for significant progress while also being exploited for malicious purposes. Chapters such as “AI Agents in Offensive Security” and “AI Agent Safety and Security Considerations” emphasize the vigilance required to mitigate risks and prevent misuse. At the same time, sections on healthcare, robotics, and the agentic economy showcase how these technologies can be leveraged to revolutionize industries, solve complex challenges, and improve human life when applied responsibly.

The future of agentic AI is both exhilarating and complex. It calls for collaboration across disciplines, a commitment to ethical governance, and a shared vision for technological progress. Academic institutions, like HKUST, will play an instrumental role in fostering the research and development needed to ensure that AI agents serve humanity’s best interests. As we stand on the cusp of this new era, works like this book are indispensable in equipping researchers, practitioners, and visionaries with the tools to navigate this rapidly evolving landscape.

I wholeheartedly recommend “Agentic AI: Theories and Practices” to anyone seeking to deepen their understanding of this fascinating field. Whether you are an academic, a technology professional, or a policymaker, this book provides a wealth of knowledge and a roadmap for engaging with agentic AI. It is an essential contribution to the discourse on how intelligent agents will shape our world.

Vice-President for Institutional Advancement,
Hong Kong University of Science and Technology
Victoria, Hong Kong

Yang Wang

Preface

This book began as an exploration into the burgeoning field of AI agents, at a time when the world was only just beginning to grasp the potential of generative models. When I first put pen to paper in the middle of 2023, the frenzy surrounding generative AI agents that we see in early 2025 was but a distant glimmer on the horizon. Now, many believe 2025 will be remembered as the year of the Agent, and for good reasons.

The capabilities of agentic AI are proving to be truly transformative. Their ability to plan, utilize tools, take actions, reflect on their performance, and manage both short and long-term memory has unlocked a new era of intelligent automation. Crucially, multi-agent systems are extending the reach of foundation models by integrating external knowledge, mitigating hallucinations, and enhancing security. These advancements are not merely incremental; they represent a fundamental shift in how we interact with and leverage artificial intelligence.

Even if the capabilities of foundational models remain static at the level of OpenAI's o3 model, which is widely recognized as the state-of-the-art as of early 2025, AI agents offer a powerful pathway to mine the full potential of this intelligence. We are at the cusp of a new scaling law for AI, one that goes beyond the traditional focus on scaling training data, model size, and computational power—a paradigm that is beginning to show diminishing returns. This new scaling law prioritizes test-time compute, process-based reward functions, and improved synthetic data generation algorithms, improved training optimization as demonstrated by DeepSeek, all aimed at fundamentally enhancing model capabilities while reducing cost.

It's important to note that this evolution in foundational models does *not* diminish the importance of AI agents. In fact, it strengthens their role. Agents can dynamically leverage real-time external data sources, providing a crucial link between the abstract knowledge of foundational models and the dynamic complexities of the real world.

This book delves into both the theoretical underpinnings and practical applications of AI agents. We explore agentic tools, frameworks, the emerging multi-agent landscape, and the nascent agentic economy. We also examine how AI agents are

poised to revolutionize a range of practical domains, from business workflows to cybersecurity, banking, insurance, healthcare, and robotics.

A critical thread woven throughout the book is the paramount importance of AI agent security and safety. We cannot afford to treat these concerns as afterthoughts. Chapter 12 is dedicated to this vital area, offering a deep dive into the challenges and considerations necessary to ensure the responsible development and deployment of these powerful technologies.

Due to the constraints of space, we were unable to cover every fascinating application of AI agents. For instance, the potential for AI agents to innovate education, or to revolutionize coding as seen in the emergence of tools like Cognition's Devin, Codium's Windsurf, and Cursor, deserve entire volumes in their own right. We have instead chosen to focus on areas where we possess deep expertise, allowing us to offer a more focused and insightful analysis.

The chapters that follow offer a comprehensive journey into the world of AI agents:

- **Chapter 1: The Genesis and Evolution of AI Agents**
- **Chapter 2: AI Agent Tools and Frameworks**
- **Chapter 3: AI Agent Ecosystem—Multi-agent Coordination**
- **Chapter 4: AI Agent Economics**
- **Chapter 5: AI Agents and Business Workflow**
- **Chapter 6: AI Agents in Offensive Security**
- **Chapter 7: AI Agents in Cyber Defense**
- **Chapter 8: AI Agents in Banking**
- **Chapter 9: AI Agents in Insurance**
- **Chapter 10: AI Agents in Healthcare Practices**
- **Chapter 11: AI Agents in Robotics**
- **Chapter 12: AI Agent Safety and Security Considerations**

You need to read this book because it's your guide to understanding and navigating the most significant technological shift since the rise of the internet: the emergence of AI agents. Here's why, in clear, direct terms:

1. **Stay Ahead of the Curve:** AI agents are moving from research labs to real-world applications at an unprecedented pace. This book gives you a crucial head start in understanding this rapidly evolving landscape, allowing you to anticipate changes, adapt your skills, and seize new opportunities.
2. **Make Informed Decisions:** Whether you're a business leader, a developer, an investor, or simply a curious individual, this book equips you with the knowledge to make informed decisions about AI agents. You'll understand their capabilities, limitations, and potential impact, enabling you to navigate the hype and make strategic choices.
3. **Gain a Competitive Edge:** Understanding AI agents will be a key differentiator in the near future. This book provides you with insights into how agents can be leveraged to improve efficiency, drive innovation, and create new value in various industries. This knowledge will give you a competitive edge in your field.

4. **Understand the Risks and Opportunities:** The book doesn't shy away from the potential risks associated with AI agents. It provides a balanced perspective, exploring both the immense opportunities and the ethical and security challenges. This understanding is crucial for responsible development and deployment of this powerful technology.
5. **Grasp the Economic Implications:** AI agents are poised to disrupt entire industries and create a new "agentic economy." This book helps you understand the economic forces at play, allowing you to anticipate shifts in the job market, business models, and value creation.
6. **Unlock the Potential of Foundation Models:** You'll learn how AI agents are the key to unlocking the full potential of even existing foundation models. This understanding will help you to maximize the value of AI investments and drive practical applications.
7. **Become a Part of the Conversation:** AI agents will increasingly shape our world. This book empowers you to become an informed participant in the crucial conversations about their ethical development, societal impact, and future trajectory.
8. **Go Beyond the Headlines:** If you want to understand the substance behind the AI agent hype, not just the sensational headlines, this book is for you. It offers in-depth analysis and insights you won't find anywhere else.

The field of AI agents is advancing at an extraordinary pace, with new innovations continuously reshaping their development and potential impact. In *Agentic AI: Theories and Practices*, we aim to lay a foundation for innovative thinking and strategic planning in this dynamic domain. However, we do not pretend to have all the answers. Instead, we embrace the complexity of the field by asking many open-ended questions—questions about AI safety and security, the influence of AI agents on economic theories, and their applications across diverse industries. These are challenges that remain unresolved and invite deeper exploration.

Our hope is that this book inspires you to think critically, approach problems creatively, and ask the kinds of questions that drive meaningful progress. As Elon Musk aptly put it, "The question is harder than the answer," referencing Douglas Adams' *The Hitchhiker's Guide to the Galaxy* to remind us that asking the right questions is often more challenging—and far more significant—than simply finding answers. We leave you with this challenge: let this book spark your curiosity and guide you in formulating the questions that will shape the future of agentic AI.

Short Recommendations

By Jim Reavis

Having spent 30+ years in cybersecurity, I found *Agentic AI: Theories and Practices* to be a refreshingly practical take on one of our industry's most pressing challenges. The book doesn't just theorize—it dives deep into the nuts and bolts of AI agent implementation, from technical architectures to the complexities of multi-agent systems. What really grabbed me was its no-nonsense approach to security and governance. As someone who's seen cybersecurity evolve from the trenches, I appreciate how this book tackles the hard questions about responsible Agentic AI deployment head-on. Whether you're in the research lab, the boardroom, or crafting policy, this is the kind of resource we desperately need to get AI security right

Jim Reavis, founder and CEO of Cloud Security Alliance

By Melan Xu

Agentic AI: Theories and Practice is an exceptional work that provides a comprehensive and practical guide to understanding the rapidly evolving field of AI agents. As the Executive Director of the World Digital Technology Academy and Secretary-General of the Cloud Security Alliance Greater China Region, I recognize the importance of bridging the gap between innovation and secure, ethical implementation.

This book stands out for its systematic approach, covering essential topics such as architecture, multi-agent systems, and their applications across industries. Its insights into governance and security considerations are particularly valuable in today's digital landscape, where trust and accountability are paramount.

I highly recommend this book to professionals, researchers, and policymakers seeking to harness the power of AI agents while navigating the complexities of our interconnected world.

Melan Xu

Executive Director, World Digital Technology Academy

Secretary-General, Cloud Security Alliance Greater China Region

By Professor Andy Chun

Ken Huang, a prolific author and renowned expert on AI and Web3, has once again delivered a masterpiece with his latest book, *Agentic AI: Theories and Practice*. This book fills a much-needed knowledge gap in the rapidly growing field of AI Agents. These agents represent the next generation in AI technology, poised to transform industries and daily life with their autonomous capabilities.

Agentic AI: Theories and Practice is an outstanding contribution to this evolving field, offering both depth and breadth in its exploration of AI agents. With over three decades of experience deploying AI systems to improve societal outcomes, I deeply appreciate the book's unique balance of theoretical rigor and real-world application.

Ken Huang provides a clear roadmap for understanding the architecture, coordination, and practical uses of AI agents across diverse industries. His insights into agentic AI's role in transforming workflows, enhancing security, and driving economic value resonate deeply with my belief in AI's potential to improve decision-making and quality of life.

I highly recommend this book as an essential guide for innovators, academics, and leaders navigating the future of AI. It's a must-read for anyone looking to stay ahead in the ever-evolving landscape of artificial intelligence.

Professor Andy Chun

Professor of Practice, The Hong Kong Polytechnic University

Distinguished Fellow, Hong Kong Computer Society

By Professor Yale Li

As the Executive Chairman of the World Digital Technology Academy and a Strategy Scientist with a global perspective on technological advancements, I am deeply impressed by the depth and breadth of *Agentic AI: Theories and Practice*. This book represents a masterful synthesis of theoretical frameworks and practical applications, guiding readers through the multifaceted world of AI agents with clarity and precision.

Its comprehensive exploration—from the architecture of AI agents to their transformative potential across industries—makes it an indispensable resource for professionals, academics, and policymakers alike. The emphasis on real-world case studies and forward-looking insights positions this work as both a reference and a roadmap for the future of agentic AI.

I highly recommend this book to anyone seeking to understand or shape the next chapter of AI-driven innovation.

Professor Yale Li
Executive Chairman, World Digital Technology Academy
Strategy Scientist, Foreign Academician of Ukrainian Academy of Engineering Sciences

By Sze Wong

It has been a privilege to review Chap. 1 of *The Genesis and Evolution of AI Agents*. Huang has taken a complex and rapidly evolving topic and presented it with clarity and simplicity, providing readers with a solid and accessible framework. This thoughtful overview serves as a strong foundation for grasping the deeper, more intricate concepts explored in later chapters.

As an investor in this space, I believe it's critical for more entrepreneurs to understand the transformative impact AI agents will have on industries and society at large. This book is an essential resource for anyone seeking to position themselves at the forefront of this groundbreaking field.

Sze Wong, Venture Investor, Tech Inventor, Zenith Venture Studio

By Chiru Bhavansikar

Chapter 3 is a must-read for any AI architect looking to design robust multi-agent systems for aligning inter-business processes. While it presumes some familiarity with AI and its frameworks, the material is well-structured and accessible to those prepared to delve into its intricacies. Even with nearly 2 years of experience working on AI agentic systems, I found the content both insightful and compelling. I can confidently attest that it provides a comprehensive and profound overview of the components necessary for building enterprise-grade AI agent systems.

Chiru Bhavansikar, Chief AI Officer, Arhasi Inc.

By Bhavin P. Kapadia

Ken Huang has once again written another comprehensive book—perfect for 2025. I was fortunate to be a reviewer for Chap. 8: *AI Agents in Banking*. Ken outlines various branches within the financial services industry that are being turned upside down by the advent of an Agentic revolution. Ken highlights specific examples where Banking is adapting to the new era of AI Agents. Readers will benefit from Ken's insights.

Bhavin P. Kapadia, Senior Advisor, AI Security Financial Services

By Akram Sheriff

Ken Huang excelled in authoring the book, especially Chap. 7 on AI Agents in Cyber Defense. According to my personal review, the chapter provides a comprehensive technical explanation, thoroughly detailing the roles and intricacies of AI agents in enhancing defensive security measures. Overall this book is a complete authority on the Topic of GenAI in Cyber Security and how different LLMs, AI agents can be used by both Offensive and defensive security Teams in the Industry to build different products.

Akram Sheriff

Senior Software Engineering Manager (GENAI, AI, Data Engineering)

Cisco Systems, San Jose, USA

By Aditya Rana

Chapter 6 covered the usage of agents for offensive security in a very detailed manner and highly recommended everyone to read it.

Aditya Rana, AquilaX.AI, Security Engineer, AI Red Team

By Dr. Tal Shapira

I had the great pleasure to review Chaps. 6, 7, and 12 which provide a comprehensive exploration of AI agents in offensive security, cyber defense, and safety considerations. Chapter 6 emphasizes the transformative potential of AI agents in offensive security, offering practical examples and code demonstrations for vulnerability discovery and adversarial simulations. Chapter 7 complements this with an in-depth look at defensive cybersecurity, focusing on multi-agent systems, incident response automation, and future trends in AI-driven defense strategies. Chapter 12 addresses the crucial topic of AI agent safety, highlighting vulnerabilities, mitigation strategies, and governance practices to ensure ethical and secure AI deployment. Together, these chapters offer valuable insights for researchers and practitioners, bridging technical innovation with the ethical imperatives of deploying AI in cybersecurity.

Dr. Tal Shapira, CTO and Co-founder, Reco AI

By Steve Wilson

The shift from creating simple chatbots to true agentic AI systems has substantial implications for development best-practices. Ken Huang, a visionary leader in advancing AI development methodologies, distills his expertise into this must-read

book. Developers will discover valuable insights to elevate their thinking and build agentic AI systems that are not only capable but also secure and responsible.

Steve Wilson

Chief Product Officer, Exabeam and Project Lead for OWASP Top 10 for LLM and Gen AI applications

Acknowledgements

I am deeply grateful to the distinguished individuals whose contributions have shaped the direction and enriched the content of this book. Their expertise, insights, and guidance have been invaluable throughout this project.

I extend my sincere gratitude to **Matthew R. Versaggi**, White House Presidential Innovation Fellow and Senior Leader in AI Healthcare, for contributing the foreword. His profound understanding of AI's transformative role in healthcare and society provides a crucial perspective on the societal challenges and opportunities AI presents. Mr. Versaggi's experience in the intersection of AI, healthcare, and quantum computing significantly enhances the book's relevance, grounding it in real-world contexts that industry professionals and researchers will deeply appreciate.

Similarly, I express my deep appreciation to **Professor Yang Wang**, Vice President for Institutional Advancement at the Hong Kong University of Science and Technology (HKUST), for his insightful foreword. Professor Wang's esteemed academic background and keen understanding of the interplay between theoretical foundations and practical innovation in AI add valuable academic depth to the introduction of this book. His endorsement of interdisciplinary collaboration and the role of academic institutions in driving breakthroughs in agentic AI enriches the book's vision of the future of the field.

This book would not have been possible without the invaluable contributions of our brilliant co-authors, whose expertise and insights have shaped its content. Each chapter reflects the dedication and effort of these individuals, and I am deeply grateful for their collaboration.

I extend my heartfelt thanks to the following contributors:

- **Jerry Huang**, AI Engineer at Google, for his technical expertise and deep understanding of AI/ML applications, which enriched multiple chapters of this book.
- **Lisa JY Tan**, CEO of Economics Design, for her thought leadership in AI agent economics and her groundbreaking perspectives on this emerging field.
- **Chris Hughes**, Co-founder and CEO of Aquia, for his contributions to the chapters on AI in offensive security, cyber defense, and safety considerations.

- **Daniel Wu**, AI/ML Leader, for his insights into the role of AI agents in the banking industry.
- **Jyoti Ponnappalli**, Vice President and Head of AI at Frontier, for her expertise in the intersection of AI agents and financial practices.
- **Grace Huang**, Product Manager at PIMCO, for her valuable contributions to the exploration of AI in financial services.
- **Bhuvaneswari Selvadurai**, CISSP, CISM, and ISACA Board Member, for her exceptional insights into AI Agent applications in insurance.
- **Krystal Jackson**, Non-Resident Research Fellow with the Center for Long-Term Cybersecurity AI Security Initiative at UC Berkeley, for her thoughtful contributions to AI safety and security considerations.

Your expertise, creativity, and dedication have been instrumental in bringing this book to life. Thank you for your invaluable contributions to advancing the conversation on AI agents and their transformative potential.

In addition to the incredible contributions from the co-authors, I would like to extend my deepest gratitude to the following individuals for their invaluable feedback, thoughtful recommendations, and detailed reviews of the whole book. Their expertise and perspectives have greatly enhanced the quality and depth of the content:

Jim Reavis

CEO and Founder, Cloud Security Alliance

Your leadership in cloud security and AI governance has been instrumental in shaping the broader discourse on securing AI systems. Your insightful feedback and support have significantly contributed to strengthening the security and governance aspects of this book.

Professor Andy Chun

Professor of Practice, The Hong Kong Polytechnic University

Distinguished Fellow, Hong Kong Computer Society

Your thoughtful analysis and recognition of this book's contribution to the field of AI agents are profoundly appreciated. Your expertise in deploying AI systems to improve societal outcomes has provided meaningful validation of the content.

Professor Yale Li

Executive Chairman, World Digital Technology Academy

Strategy Scientist, Foreign Academician of Ukrainian Academy of Engineering Sciences

Your comprehensive review of the theoretical and practical aspects of AI agents, along with your global perspective, has been instrumental in refining this book into a resource that bridges theory and application.

Steve Wilson

Lead of OWASP Top 10 for LLM Applications

Thank you for your insights and recommendations on the security considerations for AI systems, which have added depth and clarity to the discussions on safeguarding agentic AI applications.

Melan Xu

Executive Director, World Digital Technology Academy

Secretary-General, Cloud Security Alliance Greater China Region

Your exceptional review and endorsement of this book, particularly highlighting its systematic approach to AI agents and governance considerations, have added immense credibility to this work.

In addition to the reviewers who reviewed the whole book, we have individual chapter reviewers who provided detailed feedback based on their domain of expertise. Their rigorous reviews and insights have significantly contributed to improving the quality, accuracy, and clarity of the chapters:

- **Sze Wong**, for your insightful review of *The Genesis and Evolution of AI Agents*, ensuring clarity and accuracy in establishing the foundational understanding of agentic AI.
- **Chiru Bhavansikar**, for your detailed review of *AI Agent Ecosystem—Multi-Agent Coordination*, which significantly strengthened the chapter's focus on multi-agent systems and their real-world applications.
- **Anat Bremler-Barr, Tal Shapira, Akram Sheriff, Govindaraj Palanisamy, and Aditya Rana**, for your meticulous reviews and expert insights on the chapters focused on cybersecurity and defense, ensuring their relevance and technical accuracy.
- **Bhavin P. Kapadia**, for your valuable insights while reviewing *AI Agents in Banking*, helping to refine its discussion on agentic AI in financial services.
- **Madhav Chablani**, for your thoughtful review of *AI Agents in Healthcare*.

Your specialized feedback has been instrumental in elevating the quality of this book, and I deeply appreciate your dedication and contributions.

I owe immense gratitude to the editorial team at Springer, especially Jialin Yan, Lara Glueck, and Sneha Arunagiri, for their exceptional dedication, patience, and support throughout the publication process. Their hard work and guidance were invaluable, and without their contributions, this book would not have been possible. I sincerely appreciate all their efforts.

By Ken Huang, Chief Editor of this book and CEO of DistributedApps.ai

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